

1 Explainability in Reinforcement Learning

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Chapter 1

Clustering Agent Representations

As discussed in the previous chapter, current explainability methods for Reinforcement Learning (RL) still lack intrinsic approaches capable of explaining the internal mechanisms of deep neural policies. To the best of our knowledge, only [Zahavy et al., 2017] have proposed exploiting the Latent Space (LS) learned by neural policies in order to identify the concepts used by the agent. Its main idea is to analyze the Latent Representation (LR) produced by the Neural Network (NN) and organize them into clusters that correspond to high-level concepts.

While the core idea of this paper is closely aligned with our vision of explainability in RL, this approach suffers from two important limitations. First, the clusters are manually constructed. Such a procedure heavily depends on the user’s expertise and neither guarantees reproducibility or scales. Second, no evaluation protocol is proposed to assess the quality of the obtained clusters. Their relevance is essentially determined through visual inspection.

The Clustering Agent Representation (CAR) method proposed in this chapter addresses these two limitations. It automates concept discovery by relying on unsupervised clustering algorithms to identify the Concept Structure (CS) of the LS. It introduces an objective evaluation metric based on the stability of the extracted representations across independently trained policy.

This chapter is organized as follows:

- Section 1.1 lays out the theoretical assumptions underlying the proposed CAR method, which we term the *Representation Convergence Hypothesis* (Hyp. 1).
- Section 1.2 then details the CAR method itself, covering both the extraction of LR using surrogate policies and the automatic detection of CS through clustering.
- Section 1.3 introduces the metrics we propose for evaluation.
- Section 1.4 situates CAR within the broader landscape of mechanistic interpretability and outlines directions for future work.

The experimental study of CAR, covering its implementation, the environments considered, and a detailed analysis of the results, is presented separately

in Chapter ??.

1.1 Representation Convergence Hypothesis

The CAR method relies on a assumption, referred to as the *Representation Convergence Hypothesis* (Hyp. 1). This hypothesis follows the two assumptions established previously throughout this manuscript.

The *Mechanistic Hypothesis* (Hyp. ??) rejects the notion that deep NN constitute irreducible black boxes. Adopting a mechanistic perspective, we consider that every process results from a sequence of causal computations. Although these computations may be complex, they remain explainable through the analysis of the internal mechanisms.

The *Concept Hypothesis* (Hyp. ??) states that NN achieve generalization by learning concepts. During optimization, the network progressively organizes its LS into representations that capture the information required to solve the task.

Following these hypotheses, it becomes possible to explain the NN's process by understanding the organization of its LS. This family of approaches is referred to as mechanistic interpretability. In mechanistic interpretability, universality refers to the consistency with which a concept is recovered across different models or extraction processes [Templeton et al., 2024].

To provide a explanation for this phenomenon, we developed the *Representation Convergence Hypothesis* (Hyp. 1). This hypothesis states that, when NN are trained to solve the same task, a set of concepts should systematically emerge because these concepts provide a effective way of representing the problem. As a result, independently trained models should develop similar concepts in their learned representations.

Hypothesis 1: Representation Convergence Hypothesis

For a given task, NN tend to converge toward a privileged CS because it provides an efficient representation for solving the task.

It is important to clarify that this hypothesis does not imply convergence at the parameter level. Independently trained models are expected to converge toward different sets of weights. The proposed hypothesis concerns a abstract level of representation. We assume that the conceptual organization induced by their LR converges toward an equivalent CS, even though the numerical implementation of the networks differ.

The CAR method is designed to investigate this hypothesis by searching for concepts that are consistently recovered across independently trained models. In this sense, the hypothesis serves both as the foundation of the method and as the phenomenon that the method aims to investigate. If independently trained models develop common concepts for a given task, CAR should be able to recover these concepts consistently across models. Conversely, the absence of such convergence would provide evidence against the hypothesis. This also provides a stronger justification for studying these concepts, as concepts consistently recovered across different training runs are more likely to reflect properties of the task than model-specific artifacts.

The remainder of this chapter develops the methodology used to investigate this hypothesis. The *Clustering Universality* metric, introduced in Subsec-

tion 1.3.3, provides a quantitative measure of the convergence observed between the concepts identified in independently trained models.

1.2 Method

Following the work of [Zahavy et al., 2017], we analyze the LS through clustering techniques, placing our approach within the broader field of Deep Clustering. This design choice is motivated by our rejection of the *Superposition Hypothesis* (Hyp. ??) and, more generally, of any view that assumes a linearly organized CS, as discussed in Section ??.

The CAR methodology is composed of two main stages. The first stage trains multiple independent NN, called surrogate policies, which implement the same decision policy. The second stage clusters their LR in order to identify the underlying conceptual organization.

This section is organized as follows. In the first subsection, we justify the use of multiple surrogate policies in order to study the stability of representations. In the second subsection, we detail how these surrogate policies are trained so as to best match the original policy.

1.2.1 Multiple surrogate policies

The CAR method was developed to test the *Representation Convergence Hypothesis* (Hyp. 1). This requires training several NN independently and comparing their LR. Consistent similarities in the resulting CS would provide empirical support for this hypothesis.

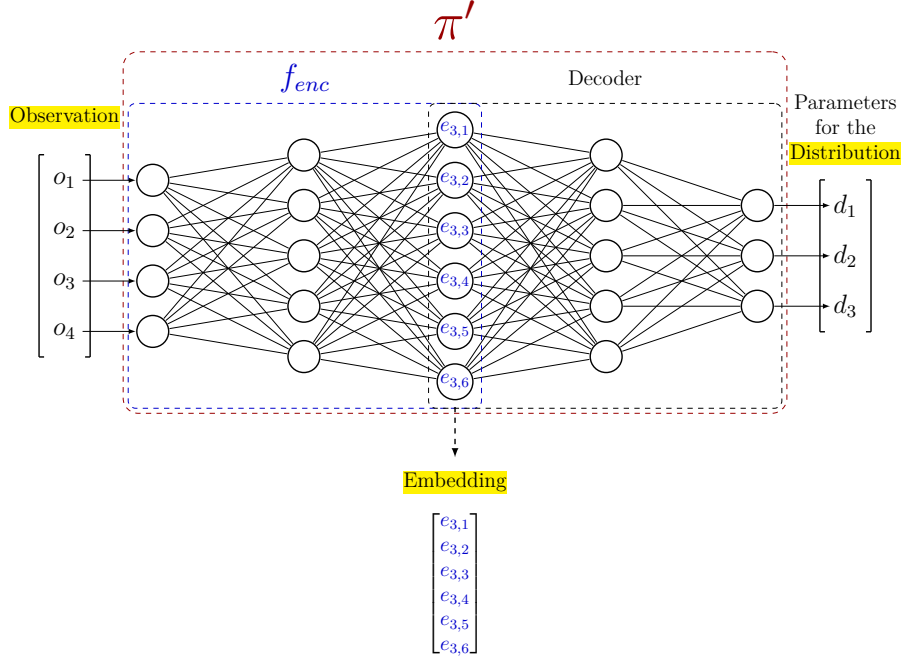
In supervised learning, this protocol is straightforward since several models can be independently trained on the same target function. RL introduces an additional difficulty. Because of stochastic exploration, independently trained agents are not guaranteed to converge toward the same policy. Even if they achieve comparable performance, they may solve the task using different strategies. This makes it difficult to determine whether differences in the LS result from different representations of the same strategy or from differences in the learned strategies themselves.

To overcome this limitation, we consider an already trained initial policy. This policy is then imitated using imitation learning [Ross et al., 2011]. In imitation learning, several surrogate NN are independently trained in a supervised manner to reproduce the behavior of the initial policy. This makes it possible to investigate whether equivalent policies also exhibit equivalent conceptual organizations.

It is important to emphasize that the use of surrogate policies does not make CAR a surrogate explainability method, as defined in Subsection ??. The surrogate policies remain deep NN. Their role is not to simplify the decision process but to provide multiple independent realizations of the same policy function, enabling the study of representation convergence.

1.2.2 Training surrogate policies

Throughout the remainder of this chapter, the initial policy we aim to explain is denoted as π . This initial policy is obtained without imposing any

Figure 1.1: Example of a surrogate policy architecture π' .

constraints on its origin. It may result from classical or deep reinforcement learning [Sutton and Barto, 2018], alpha-beta algorithm [Knuth and Moore, 1975], or even from demonstrations provided by humans [Ross and Bagnell, 2010]. The crucial requirement is the availability of a dataset containing a sufficient number of projections from the observation space $\mathcal{O}$ to the action space $\mathcal{A}$ induced by π to train a NN.

The replicated policies are referred to as surrogate policies, denoted by π' . For any observation O , the distribution over the action space produced by π' should approximate that of π as closely as possible.

The surrogate policy π' is implemented as a NN parameterized by Θ . This network is composed of two components: an encoder and a decoder, respectively highlighted by the blue and black dashed rectangles in Figure 1.1. The encoder f_{enc} maps the observation O to the embedding. The decoder, which reconstructs the action distribution parameters from the embedding.

To train π' to accurately replicate the action distribution, we define a loss function $\mathcal{L}$ optimized via gradient descent as introduced in Section ???. This loss, defined in Equation (1.1), is computed as the MSE between the parameters of the action distributions, obtained by applying the function p , which maps a distribution to its parameterization. In practice, this function does not need to be explicitly defined, as neural policies directly output the parameters of the action distribution. Once training is complete, this loss ensures that π' closely approximates the behavior of the original policy π .

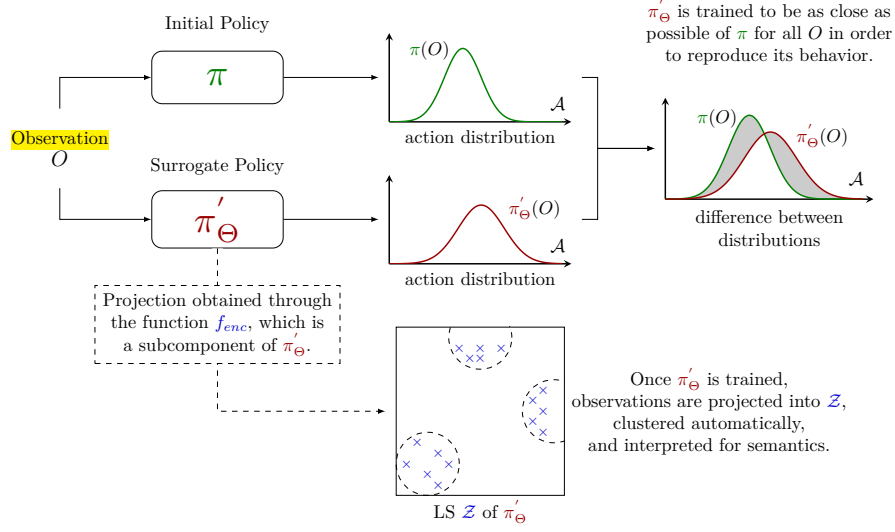


Figure 1.2: Illustration of the Clustering Agent Representation method.

$$\min_{\Theta} \sum_{O \in \mathcal{O}} \mathcal{L}(O) = \min_{\Theta} \sum_{O \in \mathcal{O}} \text{MSE}\left(p(\pi(O)), p(\pi'_\Theta(O))\right) \quad (1.1)$$

145 Once training is complete, we can identify clusters within the LS. To do so,
 146 we denote by $\mathcal{E}$ the set of embeddings obtained by applying f_{enc} to a subset
 147 of observations $\mathbb{O} \subseteq \mathcal{O}$. On the set $\mathcal{E}$, we use a clustering function f_{clust} to
 148 partition the point cloud of embedded observations into distinct clusters. Each
 149 observation is assigned a cluster label based on its proximity in the LS, resulting
 150 in the clustering $\mathcal{C}$.

151 Figure 1.2 and Algorithm 1 summarize the complete CAR procedure. Fig-
 152 ure 1.3 illustrates the analysis of a latent space obtained by applying the CAR
 153 method. The point cloud shows a two-dimensional projection of the LS of a
 154 surrogate policy trained on Space Invaders. The two-dimensional projection is
 155 used only for visualization, while the clusters are computed in the original high-
 156 dimensional LS. Colors indicate the resulting clusters and their corresponding
 157 observations. Each cluster corresponds to a distinct, human-interpretable stage
 158 of the game. The resulting organization suggests a correspondence between the
 159 position of observations in the LS and the semantic properties of the correspond-
 160 ing game states. To objectively assess this correspondence we introduce a set of
 161 evaluation metrics detailed in the following section.

Algorithm 1: Clustering Agent Representation (CAR)**Input:**Original policy π Observation dataset $\mathbb{O}$ Number of surrogate policies n Surrogate policy architecture π' (encoder f_{enc} , decoder)Learning rate η Clustering function f_{clust} **Output:**Trained surrogate policies $\pi'_{\Theta_1}, \dots, \pi'_{\Theta_n}$ Clusterings $\mathcal{C}_1, \dots, \mathcal{C}_n$ *Behavioral cloning dataset built from the target policy π* 1 $\mathcal{D} \leftarrow \{(O, p(\pi(O))) \mid O \in \mathbb{O}\};$ *MSE loss between action-distribution parameters, as defined in Equation (??)*2 $\mathcal{L} \leftarrow \text{Mean Squared Error (MSE)};$ *Independently train n surrogate policies by behavioral cloning*3 **for** $i = 1, \dots, n$ **do**4 Initialize Θ_i randomly;*Behavioral cloning of π via SGD, see Algorithm ??*5 $\Theta_i \leftarrow \text{SGD}(\mathcal{D}, \pi', \mathcal{L}, \eta, \Theta_i);$ *Cluster the latent representations of each surrogate policy*6 **for** $i = 1, \dots, n$ **do***Embeddings produced by the encoder of π'_{Θ_i}* 7 $\mathcal{E}_i \leftarrow \{f_{enc_i}(O) \mid O \in \mathbb{O}\};$ *Partition of the embedded observations into clusters*8 $\mathcal{C}_i \leftarrow f_{clust}(\mathcal{E}_i);$ 9 **Return** $(\pi'_{\Theta_1}, \dots, \pi'_{\Theta_n}), (\mathcal{C}_1, \dots, \mathcal{C}_n)$ **1.3 Evaluation Metrics**

A key contribution of this work is the development of three evaluation metrics, designed to assess the effectiveness of CAR:

- **Behavior Reconstruction** : it measures how well a surrogate policy faithfully reproduces the behavior of the original policy by comparing their cumulative rewards.
- **Clustering Universality** : it evaluates the stability of clusterings by measuring the similarity of clusters obtained from independently trained surrogate policies.
- **Abstraction Score** : it quantifies the extent to which LS clusters capture abstractions rather than directly reflecting the structure of observations or actions.

The remainder of this section details the motivation for each metric and

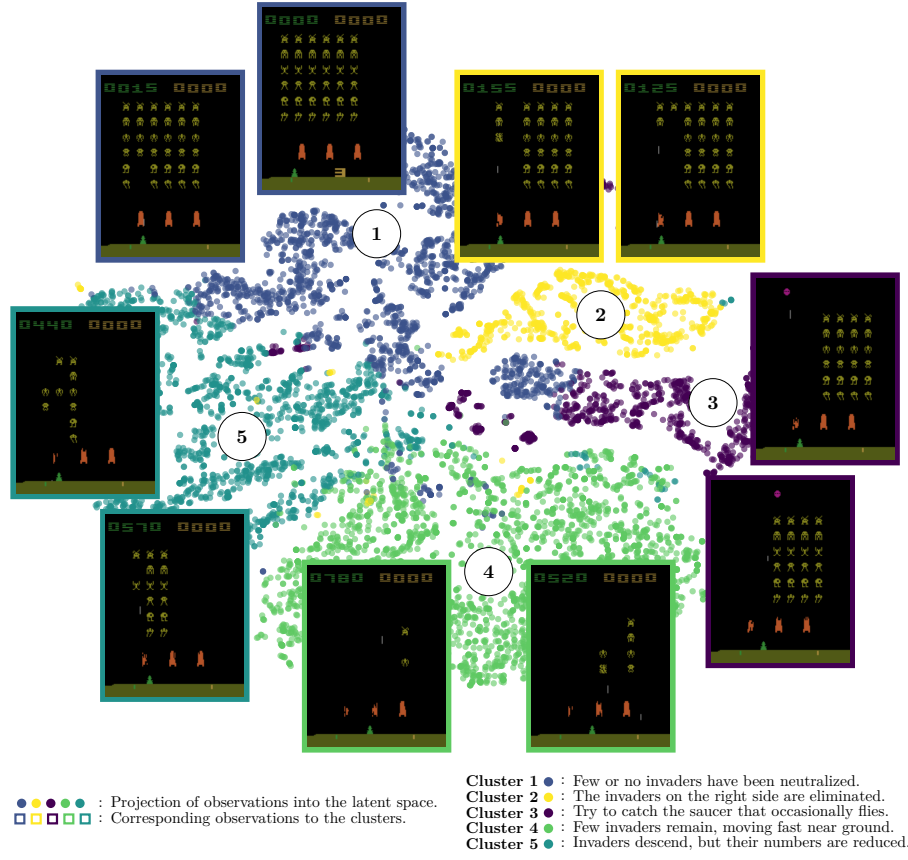


Figure 1.3: Two-dimensional projection of the LS embeddings of a surrogate policy trained on Space Invaders, together with the CAR clusters and their corresponding observations.

describes how they are calculated. In the first subsection, we develop the *Behavior Reconstruction*. We then introduce the Adjusted Rand Index (ARI) measure, which serves as the foundation for the two other metrics introduced in the subsequent subsections, namely *Clustering Universality* and *Abstraction Score*.

The following notations will be adopted for the rest of the manuscript: we consider n surrogate policies, denoted as $\pi'_1, \dots, \pi'_n$, trained as described in Section 1.2. Each surrogate policy is associated with a projection function $f_{enc_1}, \dots, f_{enc_n}$, which is applied to the set of observations $\mathbb{O}$, yielding embeddings $\mathcal{E}_1, \dots, \mathcal{E}_n$. A clustering function is then applied to each embedding set, producing partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$.

1.3.1 Behavior Reconstruction

We argue that explaining a function b that closely approximates another function a is essentially the same as explaining a . If b accurately captures the key decision patterns of a , understanding b provides meaningful insights into how a works

[Guidotti et al., 2018]. In our setting, this means that generating explanations for π' that faithfully reproduce π amounts to explaining π directly. To verify that π' is indeed a faithful copy, we introduce the *Behavior Reconstruction* metric.

This metric is computed by comparing the cumulative rewards obtained by the original policy π and the surrogate policies π' over m evaluation episodes. The behavior reconstruction metric is then defined as:

$$\text{Behavior Reconstruction} = \frac{1}{n \sum_{i=1}^m g_{\pi}^i} \sum_{j=1}^n \sum_{k=1}^m g_{\pi'_j}^k \quad (1.2)$$

In imitation learning, it is unlikely to expect the surrogate policy π' to outperform the initial policy π it aims to approximate through supervised learning [Ross and Bagnell, 2010]. Consequently this metric ranges in $[0, 1]$. Values approaching 1 indicate minimal loss in reproducing π , while values close to 0 indicate poor reproduction.

We deliberately adopt a reward-based definition rather than alternatives based on actions or action distributions (e.g., training loss, accuracy, KL divergence [Shlens, 2014], or optimal transport [Peyré and Cuturi, 2020]). Since surrogate policies are trained in a supervised manner, they cannot develop strategies beyond those demonstrated by the initial policy. This means that if the rewards achieved are comparable, the surrogate policy is effectively copying the original policy's behavior.

1.3.2 Adjusted Rand Index

Within CAR, concepts are assumed to correspond to clusters in the LS. Combined with the *Representation Convergence Hypothesis* (Hyp. 1), this leads to a direct experimental prediction: if independently trained surrogate policies converge toward the same conceptual organization, then clustering their LR should produce similar partitions.

When a particular observation is processed by the network, it is expected to activate the same concept regardless of the specific neural implementation of the policy. If independently trained policies converge toward the same conceptual organization, a given observation should activate the same concept across all models. As a consequence, observations grouped together within a cluster for one policy should also be grouped together for another independently trained policy.

In the clustering literature, the most used measures for comparing two partitions is the ARI [Hubert and Arabie, 1985]. This measure quantifies the agreement between cluster assignments independently of the spatial organization of the clusters. Both metrics introduced later in this chapter, *Clustering Universality* and *Abstraction Score*, rely on the ARI. Since the ARI is a complex measure, we review its definition and discuss its main properties.

The ARI is derived from the Rand Index (RI) [Rand, 1971], which measures the agreement between two partitions by considering every possible pair of point cloud. For each pair, four situations are possible: the point may belong to the same cluster in both partitions, to different clusters in both partitions, or they may disagree by being grouped together in only one of the two partitions. The

RI measures the proportion of pairs for which the two partitions agree:

$$\text{RI} = \frac{\text{number of agreeing pairs}}{\text{total number of pairs}}.$$

Although the RI is intuitive, it suffers from an important limitation. In most clustering problems, the majority of pairs of points belong to different clusters. Consequently, even two completely independent partitions will agree on a large number of pairs, simply because both assign them to different clusters. This effect becomes more pronounced as the number of points or clusters increases.

The ARI addresses this issue by accounting for the agreement that would be expected purely by chance. It compares the observed agreement RI to the expected agreement under random partitions $\mathbb{E}[\text{RI}]$. The score is then normalized by the maximum agreement achievable beyond this random baseline, $\max(\text{RI}) - \mathbb{E}[\text{RI}]$. As a result, an ARI of 0 corresponds to the level of agreement expected from random clusterings, while an ARI of 1 indicates perfect agreement between the two partitions:

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}$$

In practice, the ARI is computed directly from the cluster assignments of the two partitions. Let clustering A be composed of partitions $\{A_i\}$ and clustering B of partitions $\{B_j\}$. Let $a_i = |A_i|$ denote the number of samples in partition i of clustering A , and $b_j = |B_j|$ the number of samples in partition j of clustering B . Furthermore, let $n_{ij} = |A_i \cap B_j|$ be the number of samples shared by partition i of clustering A and partition j of clustering B , and let n be the total number of samples. With these definitions, the ARI can be computed as:

$$\text{ARI} = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}$$

Because it is bounded by 1, a value such as 0.5 can be misleadingly interpreted as “halfway to a perfect match.” This interpretation is incorrect because the ARI is normalized against the agreement expected from two random partitions. As a result, values close to 1 require the two clusterings to be nearly identical, while a value of 0.5 already indicates substantial agreement [Amezquita et al., 2020].

1.3.3 Clustering Universality

Having introduced the ARI, we can now define the *Clustering Universality* metric. While the ARI quantifies the similarity between two conceptual partitions, the objective of CAR is to assess the consistency of conceptual organizations across an entire set of independently trained surrogate policies.

To this end, the same set of observations is projected into the LS of each surrogate policy and clustered. Under the *Representation Convergence Hypothesis* (Hyp. 1), these independently obtained partitions are expected to be similar. *Clustering Universality* quantifies this by computing the mean pairwise ARI:

$$\text{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^n \sum_{j=0}^{i-1} \text{ARI}(C_i, C_j). \quad (1.3)$$

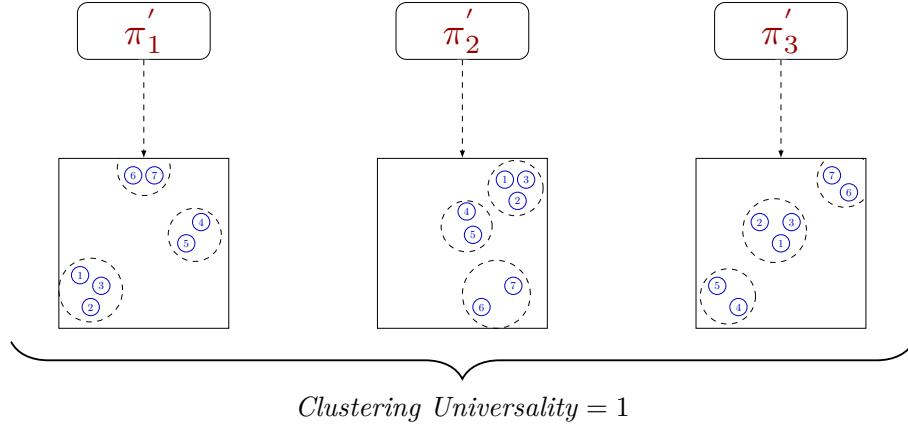


Figure 1.4: Illustration of *Clustering Universality* in the ideal case where projections of the same observations are grouped within the same clusters across all surrogate policies.

This metric takes values in $[-1, 1]$. A value close to 1 indicates that surrogate policies produce highly similar partitions, suggesting that the clusters reflect intrinsic properties of the policy. Conversely, a value near 0 implies that the similarity between partitions is no better than that of two randomly generated clusterings, meaning that the clustering fails to capture any meaningful structure in the policy. A negative value, on the other hand, signals that the agreement between partitions is worse than what would be expected by chance, indicating a systematic disagreement between the clusterings.

Figure 1.4 illustrates *Clustering Universality* in the ideal case. It shows three LS obtained by projecting the same seven observations, labeled 1 to 7, through three independently trained surrogate policies. Although the embeddings occupy different positions, the clusters group exactly the same elements in all three cases. The partitioning is therefore identical, resulting in an ARI of 1.

1.3.4 Abstraction Score

If *Clustering Universality* is high, it becomes important to determine what underlies this stability. Two possibilities arise:

- (i) the LS merely reproduces the structure of the observation space or action distribution, offering no abstraction, as defined in Definition ??;
- (ii) the LS captures similarities in decision-making processes, yielding behaviorally meaningful clusters.

To distinguish among these cases, we introduce the *Abstraction Score*. This metric contextualizes latent clusterings by examining how their patterns align with reference structures derived from the raw observation and action data. We consider : $\mathcal{C}_{obs}$, the clustering in the observation space $\mathbb{O}$, and $\mathcal{C}_{act}$, the clustering in the action-space distribution parameters. The clusterings $\mathcal{C}_{obs}$ and $\mathcal{C}_{act}$ are performed using the same algorithm, distance metric, and hyperparameters as those used for $\mathcal{C}_i$ in the LS, ensuring that any biases introduced by the clustering procedure itself are minimized.

For each latent clustering $\mathcal{C}_i$, we define its similarity to these baselines using the ARI:

$$\begin{aligned} A_i^{obs} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{obs}) \\ A_i^{act} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{act}) \end{aligned}$$

The *Abstraction Score* is subsequently calculated as one minus the average of the maximum absolute similarity values across these two reference baselines:

$$\text{Abstraction Score} = 1 - \frac{1}{n} \sum_{i=1}^n \max(|A_i^{obs}|, |A_i^{act}|) \quad (1.4)$$

The *Abstraction Score* ranges from $[0, 1]$. A score around 0 means that the latent clusters are strongly aligned with either the observation or the action space, reflecting minimal abstraction, cases (i). A score close to 1 indicates that the latent clusters are independent of both the observation and action spaces, capturing abstractions that meaningfully reflect the agent’s behavior, case (ii).

The results of applying CAR and these metrics to concrete environments are presented in Chapter ??.

1.4 Conclusion

The CAR method is the concrete instantiation of the three hypotheses developed in this manuscript: the *Mechanistic Hypothesis* (Hyp. ??), the *Concept Hypothesis* (Hyp. ??), and the *Representation Convergence Hypothesis* (Hyp. 1). It assumes that neural policies can be explained through their internal mechanisms, that these mechanisms are organized as concepts in the LS, and that independently trained policies converge toward an equivalent conceptual organization. The *Clustering Universality* and *Abstraction Score* metrics turn these assumptions into measurable quantities, which makes the hypotheses empirically testable. The experimental study reported in Chapter ?? provides support for this hypothesis-driven approach.

Beyond its role in testing these hypotheses, CAR provides a general and automated method for analysing agent representations. Both the analysis procedure and the metrics are fully automated, requiring no manual intervention or environment-specific adjustments. This makes CAR applicable to a wide range of RL settings.

A more fundamental limitation concerns the semantic interpretation of the clusters. While CAR provides a quantitative description of the structure present in the agent’s representations, the meaning assigned to individual clusters still relies on manual interpretation. CAR therefore characterizes how the LS is organized, but not what each concept stands for. Developing methods to automatically extract the semantics of each cluster is an important direction for future work.

When situating CAR within the field of explainability, substantial work remains to be done. The structure it recovers can be considered an explanation in the sense defined in Section ??, since it provides information about the concepts represented in the agent’s LS. How such explanations can be used to support validation, diagnosis, or certification, as discussed in Chapter ??,

333 remains to be established. Connecting CAR explanations more directly to the
334 agent’s decisions and failures would help determine how they can serve these
335 broader objectives.

Acronyms

ARI Adjusted Rand Index 7–11

CAR Clustering Agent Representation 1–3, 5–9, 11, 12

CS Concept Structure 1–3, 11

LR Latent Representation 1–3, 8

LS Latent Space 1–3, 5–10, 12

MSE Mean Squared Error 4, 6

NN Neural Network 1–4, 11

RI Rand Index 8, 9

RL Reinforcement Learning 1, 3, 11

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